Carlos Liang

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Experience

Bloom AI May 2024 – Present

Founding Engineer

- Collaborate directly with the founder of an early-stage AI EdTech startup to develop and deploy an AI tutoring platform serving **10,000+ students** across Australia, New Zealand, the UK and the US.
- Implemented a **RAG pipeline** with document ingestion, chunking, embeddings, and **Neo4j Aura** retrieval to ground tutor responses in course-specific learning materials.
- Integrated learning management systems through **LTI 1.3** and single sign-on, helping onboard **one major client**.
- Built AI-enabled features, including AI quiz generation, an interactive quiz builder, and a dialogue-based knowledge tracing system to track per-skill mastery and support student learning.

Macquarie Group Apr 2021 – Present

Software Engineer

- Develop software for the Macquarie Group's Futures Clearing team, working cross-functionally with business and operations to manage **US$15bn+** in client funds.
- Provide **third-level production support**, investigating production issues, identifying root causes and delivering **same-day fixes**.
- Built **FX-Deals**, automating client foreign-exchange deal generation with the implementation of a journal and authentication service with refresh tokens using Java, Spring Boot, Hibernate and AWS.
- **Led end-to-end development** of GEMS, a trade matching and allocation application designed for an estimated **100,000 daily visitors**, using TypeScript, Angular, Java, PostgreSQL, Kubernetes, Kafka and AWS.
- Mentored and led a team of new starters to release projects into production.
- Developed an AI agent on **Amazon Bedrock AgentCore** that automates interest static requests by processing Jira tickets, extracting and validating information from attachments to reduce manual effort.

Summ (formerly CryptoTaxCalculator) Jun 2020 – Mar 2021

Software Engineer

- Worked directly with founder of an early-stage cryptocurrency startup (**raised $4M**).
- Parsed cryptocurrency data and generated customer insights including tax recommendations using TypeScript.
- Helped with other ad hoc programming tasks in JavaScript and Python, including onboarding approximately 15 new crypto exchanges, web scraping and bug-fixing for clients.

Approach Solutions May 2019 – Aug 2020

Full Stack Developer

- Built over **20 crucial features** for the Growing Teams payroll app using Java, Android Studio and SAP Cloud Platform.

Nasdaq Nov 2018 – Mar 2019

Software Engineering Intern

- Automated configuration management for approximately **500 Nasdaq servers** using Puppet.
- Developed a Bash script that reduced Puppet testing time from **2 hours to 30 seconds**.

Technical Skills

Languages: Java, JavaScript, TypeScript, Python, SQL, Bash
Frameworks and Libraries: Spring Boot, Hibernate, Angular, React, Next.js, Swagger
Infrastructure & Tools: AWS, Kubernetes, Apache Kafka, Docker, Neo4j Aura, Linux, Maven, Git, PostgreSQL, IBM DB2, Splunk, Grafana
AI: Claude Code, Amazon Bedrock AgentCore

Education

University of New South Wales (UNSW Sydney) 2015 – 2020

Bachelor of Commerce (Finance) / Computer Science (Artificial Intelligence)

Projects

Personal Website | *Next.js, Three.js*

- Built a personal website using Next.js and Three.js: carlos-liang.github.io.

IFS Pal | *AI-Powered Voice Assistant*

- Developed an AI voice application using ChatGPT to facilitate real-time, interactive conversations based on Internal Family Systems (IFS) therapy.

Competitor Market Share Analysis | *Python*

- Built web scrapers for an e-commerce startup to support competitor analysis and market share modelling.